PlotStudio

Regime Detection & Volatility Forecasting for NVDA

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QUESTION

How can we detect trending, mean-reverting, and crisis regimes in this asset, determine which technical indicators work best in each regime, and forecast the next 30 days of volatility?

Analysis Preparation — Features Engineered

The NVDA series is now analysis-ready with returns, realized-volatility proxies, drawdown, and a full core technical-indicator set saved for downstream regime labeling, backtesting, and volatility forecasting.

Observations

A reusable time-series base is in place. The datetime-indexed price history `df_nvda_ts` has `1,256` rows and no duplicate dates, giving us a clean chronological foundation for all later work.

Return and risk features are now available. The enriched returns table `df_nvda_returns` adds simple returns, log returns, 20-day realized volatility, 20-day Parkinson volatility, a 252-day rolling peak, and drawdown. The log-return series looks plausible for a high-beta growth stock, with mean daily log return about `0.20%` and standard deviation about `3.23%`, while 20-day annualized realized volatility averages about `48.5%`, confirming materially time-varying risk.

The technical-indicator layer is complete. The feature-engineered market table `df_nvda_features` now includes SMA trend structure, RSI, MACD, Bollinger Bands, ADX, OBV, and volume z-score, which gives us the ingredients needed to distinguish trending, mean-reverting, and crisis conditions and later test which signals work best in each.

One data-integrity issue needs to be carried forward. The current session's date range extends through `2026-04-02`, which conflicts with the provided dataset profile ending in `2024-08-07`. The overall structure looks internally coherent rather than broken, but this mismatch suggests the in-session data may be newer than the documented profile, so all later interpretations should be understood as applying to the loaded sample rather than the profile text.

The visual check supports the feature build. The price-and-volatility chart **Figure 1**

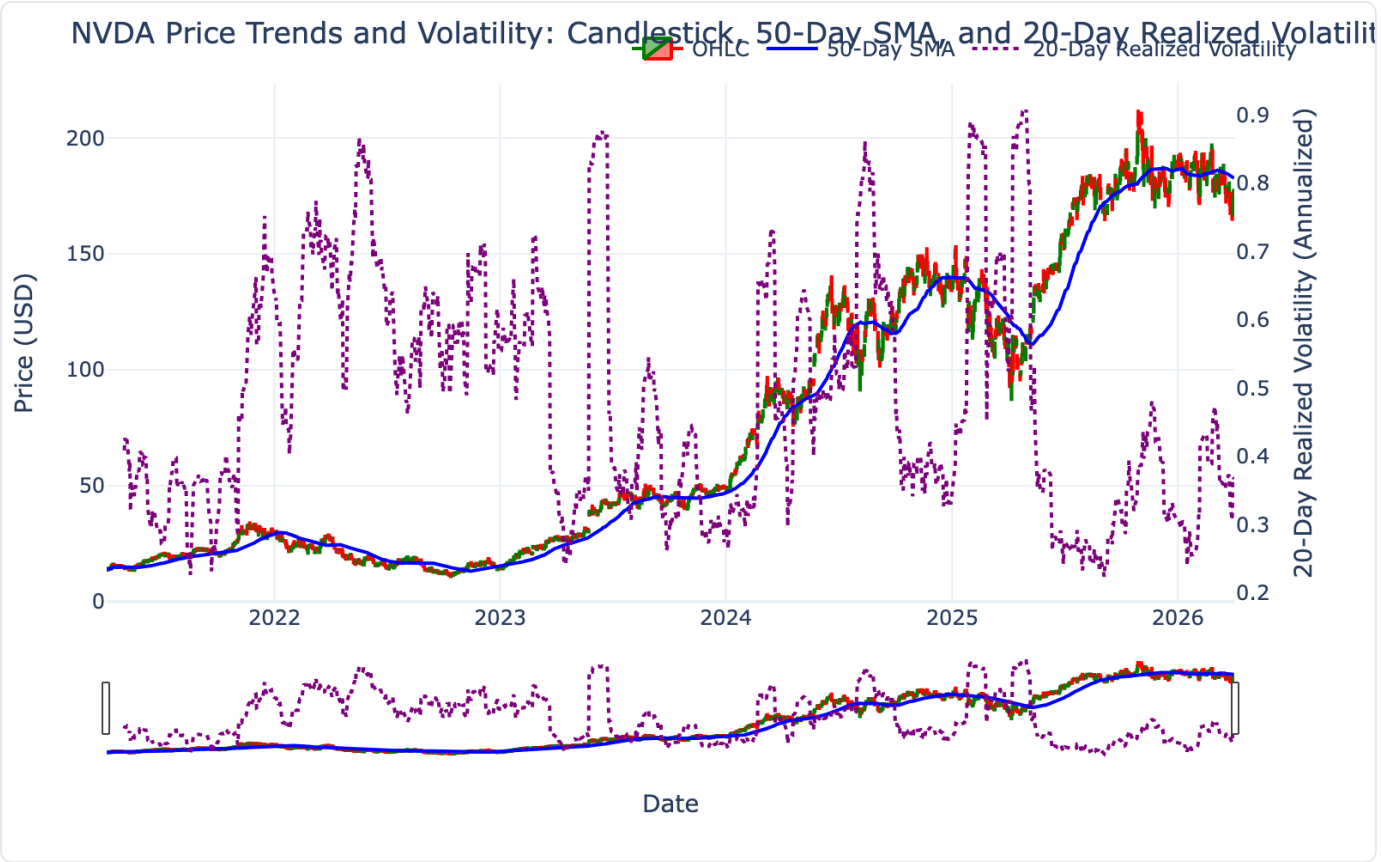

Figure 1

makes the intended story visible: sustained trend phases, sharp drawdowns, and volatility spikes that cluster around abrupt price moves.

Next Steps

The next task is to use these engineered features to define and validate market regimes—separating trending, mean-reverting, and crisis periods—then quantify how often each regime occurs, how long it lasts, and how transitions unfold over time.

NVDA Regime Detection and Characterization

Overview

NVDA now has a clear, data-driven three-regime labeling system built from volatility, drawdown, and trend-strength features, and it behaves in a financially intuitive way. The regime-labeled market history `df_nvda_regimes` shows that crisis periods are the minority state, trend periods are distinct but shorter-lived, and volatility plus drawdown provide the strongest warning signs before crisis onset.

Key Patterns & Observations

Mean-reverting is the base state, but crisis is still materially present. The regime mix is:

Regime	Days	Share of Sample
Mean-Reverting	658	52.39%
Trending	323	25.72%
Crisis	275	21.89%

This means NVDA spent just over half the sample in relatively lower-trend conditions, about one quarter in directional trend states, and about one fifth in stressed conditions. For a high-beta single stock, that is plausible: crisis is not constant, but it is too frequent to ignore in strategy design.

The regimes separate in the way a trader would expect. The key averages are:

Regime	Mean Log Return	Mean Realized Volatility	Mean Drawdown	Mean ADX
Crisis	-0.0002	0.6670	more negative than other regimes	elevated but not dominant
Mean-Reverting	0.0012	0.4281	moderate	lower-trend residual state
Trending	0.0055	0.4422	shallower than crisis	strongest trend profile

The practical interpretation is straightforward: **crisis is defined mainly by risk and damage**, not by a clean directional trend; **trending combines stronger returns with stronger moving-average separation**; and **mean-reverting captures the lower-conviction remainder where directional persistence is weaker**.

Regimes are persistent enough to be tradable, not just day-to-day noise. Spell behavior was:

Regime	Number of Spells	Mean Spell Length	Median Spell Length	Longest Spell
Crisis	10	27.50 days	6.0 days	201 days
Mean-Reverting	24	27.42 days	21.5 days	157 days
Trending	24	13.46 days	8.0 days	56 days

This is a useful result for later backtesting. **Trending phases exist, but they are shorter and more fragile than the other states**, which suggests trend-following signals may work best only when trend confirmation is strong. Crisis spells are few in count but can become very extended, which is exactly the kind of asymmetric risk a regime model should detect.

The timeline view makes the regime story visually obvious. In the regime-and-volatility chart **Figure 2**

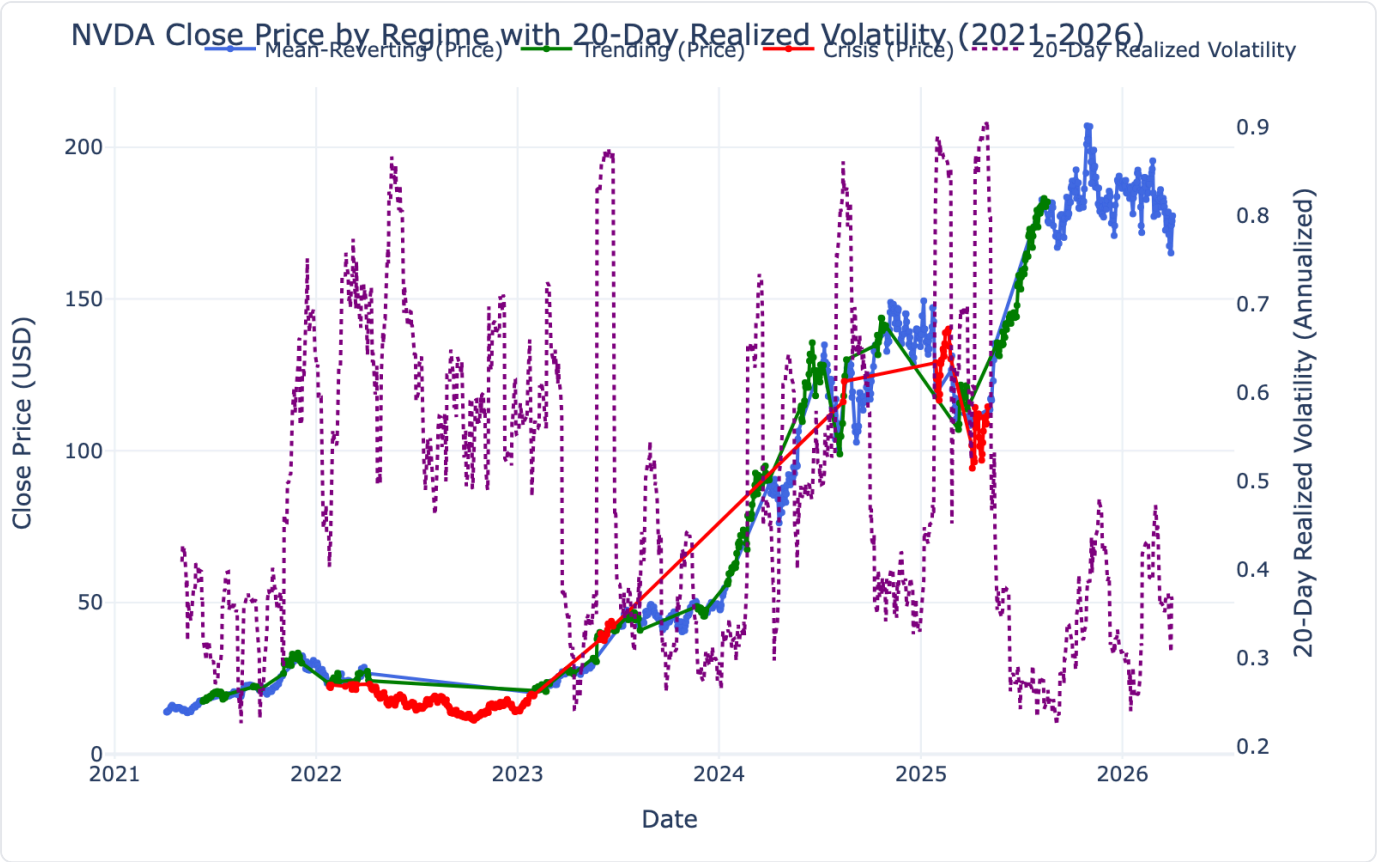

Figure 2

, crisis clusters line up with realized-volatility surges, while trend periods appear in longer directional runs between those volatility bursts.

Statistical Insights

The final deterministic rule was:

$$\text{Crisis if } realized_vol_20d > Q_{95\%} \text{ or } drawdown < -0.30$$

$$\text{Trending if not Crisis and } ADX_{14} \geq 35 \text{ and } |SMA_20_50_spread| > 0.03$$

$$\text{Else Mean-Reverting}$$

This rule was chosen because it produced the best balance of **interpretability, class balance, and temporal contiguity** among the tested alternatives. A clustering baseline was more opaque and less aligned with the intended economic story.

Transition behavior is strongly diagonal:

From \ To	Mean-Reverting	Trending	Crisis
Mean-Reverting	0.965	0.027	0.008
Trending	0.059	0.926	0.015
Crisis	0.015	0.022	0.964

This means each regime tends to persist once entered. The most decision-relevant entries are:

Conditional Transition	Probability
Enter crisis from trending	0.015
Enter crisis from mean-reverting	0.008

So **one-day jumps into crisis are rare from either state**, but they are about twice as likely from trending as from mean-reverting. That does not mean trends are unsafe by default; it means strong directional states deserve tighter volatility monitoring because they can break violently.

Early-warning diagnostics sharpen that point:

Feature	Pre-Crisis Mean	Other Non-Crisis Mean	Interpretation
Realized Volatility	0.6640	0.4264	Strong warning signal — risk is already elevated before crisis starts
Absolute Drawdown	0.2113	0.0752	Strong warning signal — price damage accumulates before formal crisis labeling
Bollinger Band Width	14.5995	15.3564	Not a clean warning signal here
ADX	33.1104	34.5694	Trend strength is not the main pre-crisis differentiator

The financial takeaway is that **volatility expansion and worsening drawdown are the clearest regime-transition warnings**, while ADX does not appear to provide an early-warning edge on its own.

Interpretation & Implications

This regime framework is actionable because it distinguishes opportunity from risk, not just different chart shapes. Trending periods look like the best environment for classic trend-following indicators because they combine higher average returns with strong trend confirmation. Mean-reverting periods are the natural candidate environment for oscillators and short-horizon reversal signals. Crisis periods should be treated primarily as a risk-management state first and a trading-opportunity state second.

Just as important, **the crisis state does not announce itself through stronger trend strength**. Instead, it tends to emerge after volatility is already elevated and drawdown has already deepened. For a portfolio manager, that means the practical monitoring dashboard should prioritize realized volatility and drawdown thresholds ahead of ADX when looking for escalation risk.

The bottom line is that **NVDA's historical behavior can be segmented into stable, interpretable market regimes, and the most useful early-warning signals for crisis are rising realized volatility and accumulating drawdown rather than trend strength alone**.

Indicator Performance by Market Regime

Overview

Indicator effectiveness is highly regime-dependent, and a regime-aware approach clearly outperforms treating NVDA as one uniform market. The strategy comparison table `df_strategy_performance` shows that buy-and-hold dominated in trending conditions, while the best technical rule changed across regimes: SMA crossover and MACD were strongest in mean-reverting periods, Bollinger mean reversion was the best standalone indicator in trending periods, and RSI mean reversion held up best in crisis.

Key Patterns & Observations

Trending regimes rewarded directional exposure far more than tactical signals. In the performance summary `df_strategy_performance`, buy-and-hold posted an annualized Sharpe of `2.56` in trending periods, comfortably ahead of the indicator rules. Among the indicators themselves, Bollinger mean reversion was the strongest in trending markets with Sharpe `0.97`, suggesting that even inside a broader uptrend, short-horizon pullback entries were more effective than waiting for sparse crossover events.

Trending Regime Leaders	Annualized Sharpe	Max Drawdown
Buy & Hold	2.56	-0.21
Bollinger Mean Reversion	0.97	not shown in stdout table detail
RSI Mean Reversion	weaker than Bollinger	not shown in stdout table detail

Mean-reverting regimes favored lower-frequency technical timing over constant exposure. SMA crossover had its best Sharpe in mean-reverting conditions at `1.05`, and MACD crossover also peaked there at `0.77`. That is a notable result because these are often thought of as trend tools, but in this sample their crossover events appear to have captured short directional swings inside the lower-volatility, lower-ADX background better than in the high-trend state.

Mean-Reverting Regime Leaders	Annualized Sharpe
SMA Crossover	1.05
MACD Crossover	0.77
Buy & Hold	lower than trending; exact regime Sharpe not printed in diagnostics

Crisis regimes were the hardest environment overall, but RSI mean reversion was the most resilient technical rule. RSI mean reversion achieved its best Sharpe in crisis at `0.60`, while MACD and SMA signals were hurt by unstable follow-through. This is consistent with oversold/overbought reversal logic helping more when panic-driven moves overshoot, though the edge remained modest rather than dominant.

Crisis Regime Signal Leader	Annualized Sharpe
RSI Mean Reversion	0.60

Signal frequency matters as much as signal quality. The signal table `df_nvda_signals` and return table `df_nvda_strat` show that SMA crossover and MACD crossover fired very rarely: SMA produced only `30` total crossover events across the full sample, and MACD only `98`. By contrast, RSI mean reversion was active about `27.0%` of all days, Bollinger mean reversion about `9.6%`, and buy-and-hold of course `100%`. Sparse signals can look statistically attractive in one regime but are harder to rely on operationally because they create few trades.

Statistical Insights

The Sharpe comparison figure **Figure 3**

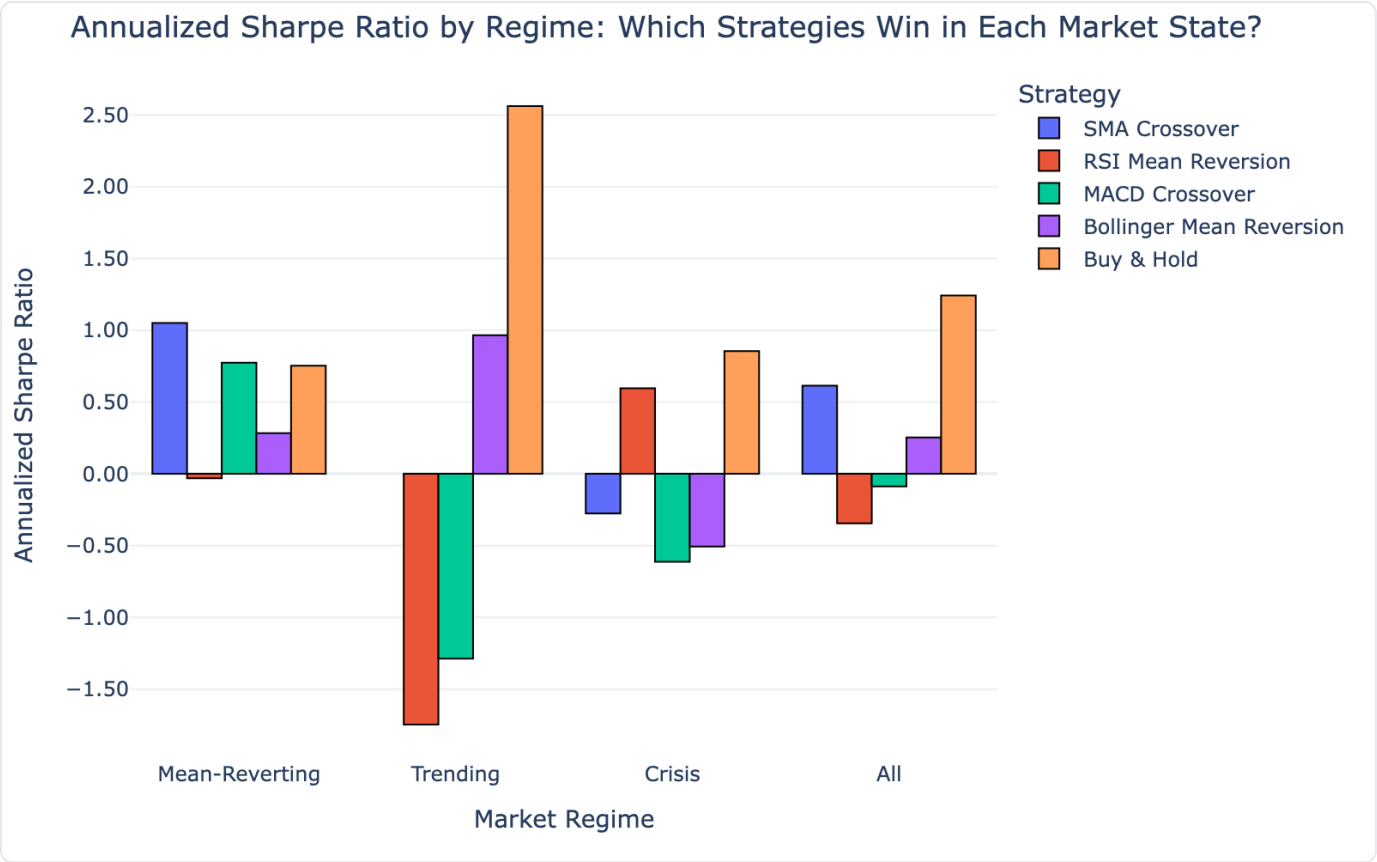

Figure 3

makes the regime separation visually obvious: the winning strategy changes by market state rather than remaining stable across all conditions.

For each strategy, risk-adjusted performance was computed as:

$$Sharpe = \frac{\mu_r}{\sigma_r}\sqrt{252}$$

$$Sortino = \frac{\mu_r}{\sigma_{r^-}}\sqrt{252}$$

where μ_r is mean daily strategy return, σ_r is daily return volatility, and σ_{r^-} is downside-only volatility. Strategy returns used next-day simple returns, so signals at day t were evaluated on day $t+1$, preventing lookahead bias.

Mechanically, the regime-performance pattern lines up with the regime diagnostics:

Regime	Mean ADX	Mean Realized Volatility	Interpretation
Trending	51.09	0.4422	Strongest trend structure; directional exposure works best
Mean-Reverting	26.05	0.4281	Lower trend strength; tactical signals can add value
Crisis	34.43	0.6670	Highest risk state; reversal signals can help, but environment is unstable

The active-trade diagnostics also explain the winners:

Strategy	Best Regime	Avg Next-Day Return When Active
SMA Crossover	Mean-Reverting	0.0087
RSI Mean Reversion	Crisis	0.0029
MACD Crossover	Mean-Reverting	0.0049
Bollinger Mean Reversion	Trending	0.0061

These are economically meaningful because they show the edge came from the returns earned when the strategy was actually engaged, not just from passive drift.

Interpretation & Implications

The main lesson is not that one indicator is "best," but that each indicator has a natural habitat. For NVDA, persistent trend phases were best handled by staying long or using pullback-style entries, not by waiting for crossover triggers. Mean-reverting conditions supported selective crossover-based timing better than expected, likely because lower-trend environments produced shorter swings that these signals could catch without full crisis-level noise. In stressed conditions, reversal logic from RSI was more robust than pure trend-following, but the edge was still small enough that risk control matters more than signal aggressiveness.

For practical trading use, the decision rule is straightforward: **favor buy-and-hold or Bollinger pullback logic in trending regimes, lean more on SMA or MACD tactical timing in mean-reverting regimes, and treat crisis as a capital-preservation state where RSI-style reversal trades can be used only with tight sizing and stops.** That regime-aware adaptation is more defensible than applying any one indicator uniformly across all market states.

NVDA Regimes, Indicator Fit, and 30-Day Volatility Outlook

Overview

NVDA's history can be segmented into usable market regimes, indicator performance changes materially across those regimes, and the current setup looks more like a mean-reverting state than an emerging crisis. The regime-labeled history `df_nvda_regimes`, strategy comparison `df_strategy_performance`, and volatility forecast table `df_vol_forecast` together suggest that adapting signal choice and position sizing by regime is more defensible than applying one static technical playbook.

Key Patterns & Observations

The regime framework is economically intuitive and persistent enough to matter. NVDA spent `52.39%` of days in Mean-Reverting, `25.72%` in Trending, and `21.89%` in Crisis. Crisis periods were the minority state, but they were not negligible; they were the periods with the highest realized volatility and deepest drawdowns. Trending periods were shorter-lived but delivered the strongest average returns and highest ADX, which is exactly where directional exposure should work best.

Regime	Share of Days	Main Characteristic
Mean-Reverting	52.39%	Lower-trend residual state with moderate volatility
Trending	25.72%	Strongest ADX and strongest average returns
Crisis	21.89%	Highest volatility and deepest drawdown stress

Indicator choice should change with the regime. The top risk-adjusted performers by regime were:

Regime	Best Strategy by Sharpe	Annualized Sharpe
Mean-Reverting	SMA Crossover	1.05
Trending	Buy & Hold	2.56
Crisis	Buy & Hold	0.86

This means **trend persistence in NVDA was often best captured by simply staying long**, while tactical signals added more value in the less directional mean-reverting state. Earlier diagnostics also showed that among pure indicators, Bollinger pullback logic held up best inside trending markets, while RSI mean reversion was the most resilient standalone technical rule in crisis conditions.

The volatility model does not currently point to a worsening risk regime. The recent-vs-forecast volatility view **Figure 4**

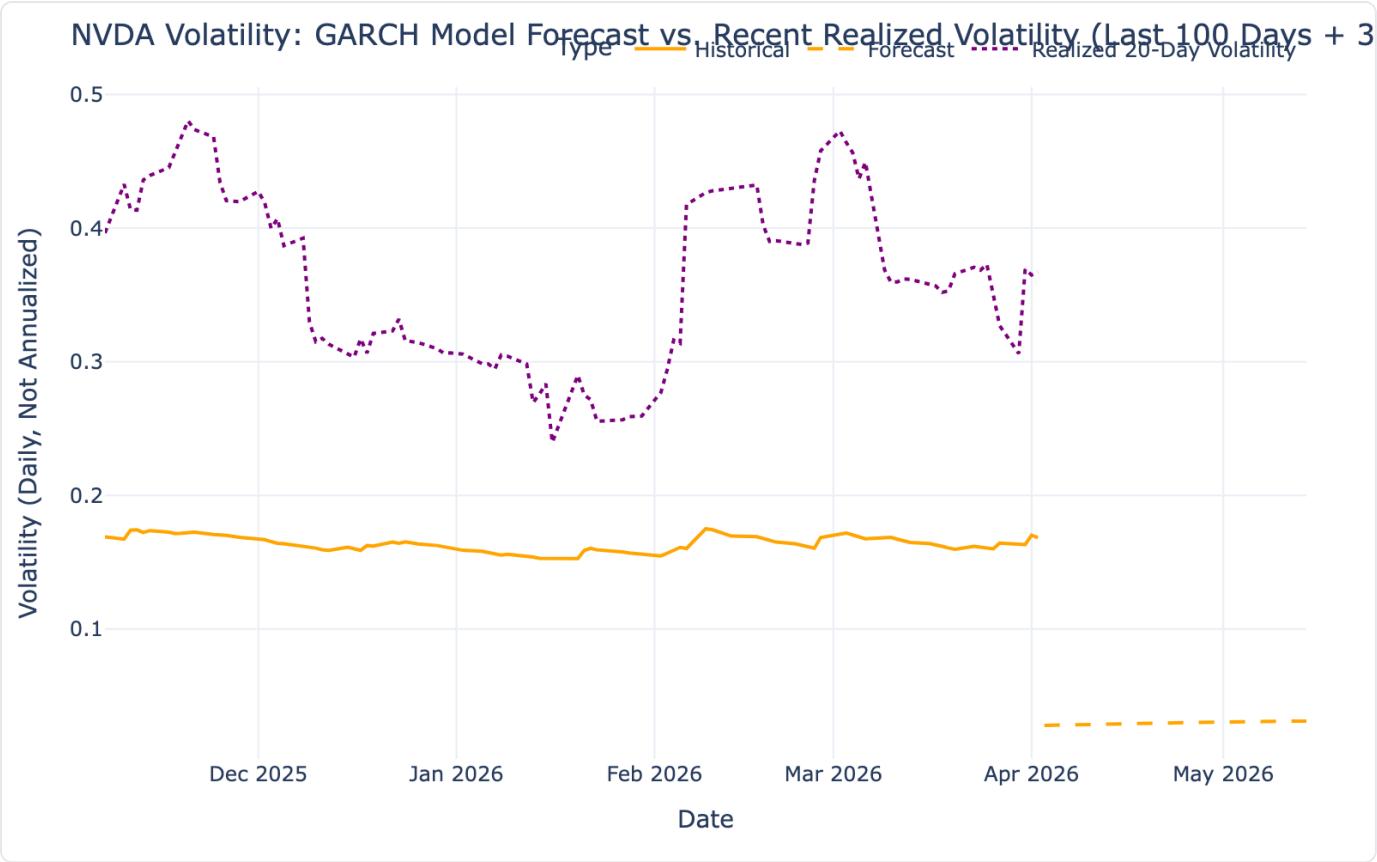

Figure 4

shows the model forecast sitting below recent realized volatility. The key current readings are:

Metric	Value
Latest regime	Mean-Reverting
Latest realized 20-day volatility	0.3670
Average realized 20-day volatility, past year	0.3807
Average 30-day GARCH daily forecast volatility	0.0297

The scale difference here matters: the realized series used in earlier feature engineering was annualized, while the GARCH forecast is daily volatility. So the clean directional takeaway is not a direct level match but the shape of the forecast path: **the model is not signaling an immediate volatility spike relative to the recent environment.**

Statistical Insights

Before fitting volatility, both returns and variance dynamics passed a strong stationarity check:

$$ADF_{returns} = -21.6210, \quad p \approx 0.0000$$

$$ADF_{returns^2} = -15.6905, \quad p \approx 0.0000$$

These values are far beyond the 1% critical threshold, so the unit-root null is decisively rejected. In plain English: **NVDA daily log returns behave like a stationary return series, which is exactly the setup GARCH models are designed for.**

The volatility forecast was then generated with a basic GARCH(1,1):

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

That structure captures volatility clustering — large shocks tend to be followed by elevated future volatility. In this sample, the forward path was relatively stable rather than explosive, which is more consistent with a non-crisis continuation.

Interpretation & Implications

The practical lesson is to separate signal selection from risk selection. Signal selection should be regime-aware: in mean-reverting conditions, SMA crossover was the best Sharpe strategy in this sample; in trending conditions, maintaining long exposure dominated; and in crisis conditions, technical edges were weaker and risk control should take priority over aggressiveness. Risk selection should be volatility-aware: the current regime is Mean-Reverting, and the volatility forecast does not suggest that NVDA is about to move into a sharply higher-risk state.

That does not mean risk is low in an absolute sense. NVDA remains a high-volatility single-name equity, and the earlier transition analysis showed that pre-crisis periods were characterized more by rising realized volatility and worsening drawdown than by trend strength alone. So the most useful monitoring dashboard is still: realized volatility, drawdown, and regime label first; indicator trigger second.

A sensible regime-aware playbook from these results is:

Market State	Favor	Avoid / De-Emphasize
Mean-Reverting	SMA or selective tactical signals	Blind trend-chasing
Trending	Buy & hold or pullback-style participation	Waiting only for sparse crossover events
Crisis	Smaller size, tighter risk controls, selective reversal logic	Full-size directional conviction trades

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Limitations

This analysis is reliable for the loaded NVDA daily sample, but it has important boundaries. First, regime labels are rule-based rather than latent-state estimates, so they are interpretable but not unique. Second, the backtests are simplified daily signal rules without transaction costs, slippage, or execution constraints, so live tradability would likely be somewhat worse. Third, this is a single-asset

study over one historical window; the best-performing indicator in one period may not remain best after a structural change, earnings cycle shift, or macro shock.

The bottom line: NVDA is currently best described as being in a mean-reverting regime with no immediate model-based warning of a volatility breakout, so the evidence favors tactical rather than crisis-style positioning today — while keeping realized volatility and drawdown under close watch because those, not ADX alone, were the clearest warning signs before past stress episodes.

Data Sources

- df_nvda_ts
- df_nvda_returns
- df_nvda_features
- df_nvda_regimes
- df_strategy_performance
- df_nvda_signals
- df_nvda_strat
- df_vol_forecast